

Closing the order law’s $\tau = 0$ gap: an SYT with all-free descents exists, via a descent-set Kostka number

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Abstract

The order law states $\text{ord}_{x=q^2} Z_\lambda = \min_{T \in \text{SYT}(\lambda)} s(T) = \tau(\tau + 1)/2$ for a parity-twisted descent statistic s . Its lower bound and the $\tau > 0$ achievability were proved earlier; the lone remaining gap was $\tau = 0$ achievability — the existence of a standard Young tableau with no “paid” descent. We close it: such a tableau exists for every $\tau = 0$ shape, and in fact the number of them is a Kostka number $K_{\lambda, \mu}$, positive exactly when $\tau = 0$. The proof is the classical descent-set/Kostka identity plus a two-line dominance computation; we also correct an earlier mis-attribution of this gap to Swanson’s modular major-index existence theorem.

1 Setup

Fix $n \geq 1$ and a partition $\lambda \vdash n$ with $\ell = \lambda'_1$ rows (the length of the first column). For a standard Young tableau $T \in \text{SYT}(\lambda)$, the *descent set* is

$$\text{Des}(T) = \{ i \in [1, n-1] : i+1 \text{ lies in a strictly lower row than } i \text{ in } T \}.$$

Define the *parity-twisted weight*

$$w_i = \begin{cases} 2i-1, & n-i \text{ odd,} \\ 0, & n-i \text{ even,} \end{cases} \quad s(T) = \sum_{i \in \text{Des}(T)} w_i.$$

Call a position i *free* if $n-i$ is even (equivalently $i \equiv n \pmod{2}$), so $w_i = 0$) and *paid* otherwise. Let

$$S = \{ i \in [1, n-1] : i \equiv n \pmod{2} \}$$

be the set of free positions. Set

$$\tau = \max(0, 2\ell - n - 1).$$

Since the paid positions are exactly those carrying a nonzero weight, we have the basic equivalence

$$s(T) = 0 \iff \text{Des}(T) \subseteq S \quad (\text{all descents free}). \quad (1)$$

For background (established in the companion note [2026-05-31-orderlaw-descent-minimum](#)): the lower bound $s(T) \geq \tau(\tau + 1)/2$ holds for all $T \in \text{SYT}(\lambda)$, and for $\tau > 0$ there is an explicit minimizer attaining equality. The remaining gap was the case $\tau = 0$: does there exist T with $s(T) = 0$? By (1) this is exactly the question of whether some $T \in \text{SYT}(\lambda)$ has all of its descents at free positions. We answer it affirmatively and, in fact, count such tableaux.

2 The descent-set Kostka identity

Lemma 1 (classical). *For any subset $S = \{s_1 < \dots < s_k\} \subseteq [1, n - 1]$, let*

$$\alpha(S) = (s_1, s_2 - s_1, \dots, s_k - s_{k-1}, n - s_k)$$

be the composition of n whose set of partial sums (excluding the full sum n) is exactly S . Then

$$\#\{T \in \text{SYT}(\lambda) : \text{Des}(T) \subseteq S\} = K_{\lambda, \alpha(S)},$$

the number of semistandard Young tableaux of shape λ and content $\alpha(S)$, equivalently $\langle s_\lambda, h_{\alpha(S)} \rangle$. Since $K_{\lambda, \alpha}$ depends only on the rearrangement of α into a partition, write $\mu = \text{sort}(\alpha(S))$; then the count is $K_{\lambda, \mu}$.

Proof. This is the standard descent-set/Kostka identity; see Stanley, *Enumerative Combinatorics* vol. 2, Cor. 7.19.5 and Thm. 7.19.7, and Gessel. Concretely it is the pairing of the complete-homogeneous expansion $h_\alpha = \sum_\lambda K_{\lambda\alpha} s_\lambda$ against the fundamental quasisymmetric expansion $s_\lambda = \sum_{T \in \text{SYT}(\lambda)} F_{\text{Des}(T)}$, together with the fact that the coefficient extracting $\text{Des}(T) \subseteq S$ from $F_{\text{Des}(T)}$ recovers precisely the tableaux whose descent set is contained in S . $\square$

3 The composition for free positions

Lemma 2. For $S = \{i \in [1, n-1] : i \equiv n \pmod{2}\}$:

- if n is even: $S = \{2, 4, \dots, n-2\}$, so $\alpha(S) = (2, 2, \dots, 2)$ with $n/2$ twos, and $\mu = (2^{n/2})$;
- if n is odd: $S = \{1, 3, \dots, n-2\}$, so $\alpha(S) = (1, 2, 2, \dots, 2)$ (a single 1 followed by $(n-1)/2$ twos), and $\mu = (2^{(n-1)/2}, 1)$.

In both cases the conjugate partition is

$$\mu' = (\lceil n/2 \rceil, \lfloor n/2 \rfloor),$$

a partition with at most two parts; in particular $\mu'_1 = \lceil n/2 \rceil$.

Proof. We read off $\alpha(S)$ from the partial-sum description of Lemma 1.

n even. The free positions are the even integers in $[1, n-1]$, namely $S = \{2, 4, \dots, n-2\}$. The consecutive gaps are all 2, and the final gap is $n - (n-2) = 2$, so $\alpha(S) = (2, 2, \dots, 2)$ with $n/2$ parts. Its partial sums are $2, 4, \dots, n$, and intersecting with $[1, n-1]$ recovers S , as required. Sorting gives $\mu = (2^{n/2})$.

n odd. The free positions are the odd integers in $[1, n-1]$, namely $S = \{1, 3, \dots, n-2\}$. The first gap is 1, the remaining consecutive gaps are 2, and the final gap is $n - (n-2) = 2$, so $\alpha(S) = (1, 2, 2, \dots, 2)$ with a leading 1 and $(n-1)/2$ twos. Its partial sums are $1, 3, 5, \dots, n$, whose intersection with $[1, n-1]$ is S . Sorting gives $\mu = (2^{(n-1)/2}, 1)$.

For the conjugate, count column lengths. In either case μ has parts equal to 2 and possibly one part equal to 1. The first column of μ has one box per part, i.e. μ'_1 equals the number of parts: $n/2$ when n is even and $(n-1)/2 + 1 = (n+1)/2$ when n is odd; both equal $\lceil n/2 \rceil$. The second column has one box for each part of size ≥ 2 , i.e. μ'_2 equals the number of twos: $n/2$ when n is even and $(n-1)/2$ when n is odd; both equal $\lfloor n/2 \rfloor$. There is no third column. Hence $\mu' = (\lceil n/2 \rceil, \lfloor n/2 \rfloor)$. $\square$

4 Dominance

We write $\supseteq$ for the dominance order on partitions of n .

Lemma 3. With μ as in Lemma 2,

$$K_{\lambda, \mu} > 0 \iff \lambda \supseteq \mu \iff \mu' \supseteq \lambda' \iff \lambda'_1 \leq \lceil n/2 \rceil \iff \tau(\lambda) = 0.$$

Proof. The first equivalence is the Kostka positivity criterion: $K_{\lambda\mu} > 0$ if and only if λ dominates μ . The second is the standard fact that dominance is reversed under conjugation: $\lambda \supseteq \mu \iff \mu' \supseteq \lambda'$.

For the third equivalence, recall from Lemma 2 that $\mu' = (\lceil n/2 \rceil, \lfloor n/2 \rfloor)$ has only two (possibly nonzero) parts. The dominance condition $\mu' \supseteq \lambda'$ requires

$$\sum_{i \leq j} \mu'_i \geq \sum_{i \leq j} \lambda'_i \quad \text{for all } j \geq 1.$$

For $j = 1$ this reads $\lceil n/2 \rceil \geq \lambda'_1$. For $j \geq 2$ the left-hand side is already the full sum $\mu'_1 + \mu'_2 = \lceil n/2 \rceil + \lfloor n/2 \rfloor = n$, while the right-hand side $\sum_{i \leq j} \lambda'_i \leq n$; the inequality is therefore automatic. Hence the single binding constraint is $\lambda'_1 \leq \lceil n/2 \rceil$, i.e. $\ell \leq \lceil n/2 \rceil$.

Finally, $\lambda'_1 \leq \lceil n/2 \rceil \iff 2\lambda'_1 \leq 2\lceil n/2 \rceil$. Since $2\lceil n/2 \rceil \leq n + 1$ for both parities (equality when n is odd, and $2\lceil n/2 \rceil = n \leq n + 1$ when n is even), and since λ'_1 is an integer, $\lambda'_1 \leq \lceil n/2 \rceil$ is equivalent to $2\lambda'_1 \leq n + 1$, i.e. to $2\lambda'_1 - n - 1 \leq 0$, i.e. to $\tau = \max(0, 2\lambda'_1 - n - 1) = 0$. $\square$

5 Main theorem

Theorem 1. *Let $\lambda \vdash n$ with $\tau(\lambda) = 0$, and let μ be as in Lemma 2. Then*

$$\#\{T \in \text{SYT}(\lambda) : s(T) = 0\} = K_{\lambda,\mu} \geq 1.$$

In particular there exists $T \in \text{SYT}(\lambda)$ achieving $s(T) = 0 = \tau(\tau + 1)/2$. Combined with the lower bound $s(T) \geq \tau(\tau + 1)/2$ and the explicit $\tau > 0$ construction, the order law

$$\min_{T \in \text{SYT}(\lambda)} s(T) = \frac{\tau(\tau+1)}{2}$$

now holds, with full rigor, uniformly over all partitions λ .

Proof. By (1), $s(T) = 0$ if and only if $\text{Des}(T) \subseteq S$. Applying Lemma 1 with this S counts such tableaux as $K_{\lambda, \alpha(S)} = K_{\lambda, \mu}$, where Lemma 2 identifies μ . Since $\tau(\lambda) = 0$, Lemma 3 gives $K_{\lambda, \mu} > 0$, hence the count is at least 1 and a minimizer with $s(T) = 0$ exists. The remaining clause assembles the three legs: the lower bound and $\tau > 0$ achievability are established in the companion note, and the present theorem supplies $\tau = 0$ achievability, so the minimum equals $\tau(\tau + 1)/2$ for every λ . $\square$

6 Remarks

Remark 1 (Exact count of minimizers). For $\tau = 0$ the number of order-law minimizers is precisely the Kostka number $K_{\lambda,\mu}$ — a refinement of mere existence. For instance, with $n = 7$: the shape $\lambda = (4, 2, 1)$ gives $K_{\lambda,\mu} = 6$, and $\lambda = (5, 2)$ gives $K_{\lambda,\mu} = 5$.

Remark 2 (Conjugate form). Under transposition one has $s(T') = \binom{n}{2} - s(T)$. Consequently the count above can be read on the conjugate shape as

$$\#\{T' \in \text{SYT}(\lambda') : \text{every paid position is a descent}\} = K_{\lambda,\mu}.$$

Remark 3 (Correction of attribution). This gap was previously conjectured to coincide with Swanson’s existence theorem for tableaux with a prescribed modular major index (arXiv:1701.04963). It does not. Swanson controls

$$\text{maj}(T) = \sum_{i \in \text{Des}(T)} i \pmod{n},$$

a single linear functional of the descent set, reduced modulo n . The condition $s(T) = 0$ instead constrains *which positions, by parity*, may be descents. The latter is a finer, positional condition on the descent set that Swanson’s mod- n machinery cannot detect: a tableau can realize any target value of $\text{maj} \pmod{n}$ while still placing a descent at a wrong-parity (paid) position. The correct and far more elementary tool is the descent-set Kostka positivity of Lemma 1.

7 Computational verification

All three legs were verified by an independent script `verify_kostka_bridge.py`, which enumerates semistandard tableaux for the Kostka side using no shared code with the SYT/ $s(T)$ side:

- (i) $\#\{T : s(T) = 0\} = K_{\lambda,\mu}$: **0 mismatches** over all 194 partitions with $n \leq 11$.
- (ii) $\lambda \trianglerighteq \mu \iff \tau = 0$: **0 failures** over all 507 partitions with $n \leq 14$.
- (iii) the single binding constraint $\ell \leq \lceil n/2 \rceil$: **0 failures** for $n \leq 12$.

Earlier, all $\tau = 0$ shapes with $n \leq 12$ were brute-confirmed to admit an explicit $s(T) = 0$ witness, in agreement with the existence statement of Theorem 1.